Srishti Shree Nidhi

Computer Science Undergraduate | Full-Stack and Machine Learning
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SUMMARY

Final-year B.Tech Computer Science student who builds full-stack and AI/ML applications end to end: a MERN + RAG platform, an LLM maintenance copilot (RAG, tool calling, grounded report generation), computer vision systems and NLP pipelines. Strong in Python, machine learning and LLM application development. Seeking Software Engineer or ML Engineer roles.

EDUCATION

RV University, Bengaluru | B.Tech in Computer Science and Engineering | 2023 – 2027 (Expected)
CGPA: 7.48 / 10

Lord Buddha Public School, Kota | Class XII (CBSE) | 2022 | Percentage: 70.20%
Lucent International School, Dehradun | Class X (CBSE) | 2020 | Percentage: 82.30%

TECHNICAL SKILLS

- **Languages:** Python, Java, SQL, HTML, CSS
- **Web and Backend:** React, Node.js, Express, Flask, FastAPI, REST APIs, Socket.io, JWT authentication
- **Machine Learning:** scikit-learn, TensorFlow, Keras, PyTorch, Pandas, NumPy
- **Computer Vision:** OpenCV, YOLOv8, Tesseract OCR
- **NLP and LLMs:** LLMs, Prompt Engineering, RAG, LangChain, Agentic AI, Generative AI, spaCy, Transformers, sentence-transformers
- **Databases and Tools:** MongoDB, SQL, Git, GitHub, Git workflows, Docker, CI/CD, VS Code, Streamlit, Google Colab
- **Core CS:** Data Structures and Algorithms, Object-Oriented Programming, System Design

PROJECTS

AeroVision – AI Aircraft Defect Detection and Maintenance Copilot | Aug 2026 – Oct 2026
Tech: Python, OpenCV, YOLOv8, PyTorch, Groq Llama 3.3 70B, sentence-transformers, Pydantic, Streamlit
- Built a Streamlit system that detects and classifies four aircraft-surface defect types (cracks, corrosion, dents, rivet damage) using OpenCV with an optional YOLOv8 backend, adding severity scoring, explainable heatmaps and scan-to-scan progression tracking.
- Trained a PyTorch LSTM for engine remaining-useful-life prediction on NASA C-MAPSS data and fused it with the defect findings into one structured inspection record.
- Built an LLM maintenance copilot: a RAG knowledge base of repair guidance (sentence-transformers + FAISS, filtered by defect type and severity) and a tool-calling agent (Groq Llama 3.3 70B with Gemini fallback) that runs the detector, severity, progression and RUL functions to answer questions such as "is this scan worse than the last one?"
- Added grounded report generation: LLM output is validated with Pydantic and checked against pipeline results, every recommendation must cite a retrieved guidance chunk, and failures fall back to a template; retrieval quality is measured with hit@k and MRR.

VentureAI – Startup–Investor Matching Platform | Jan 2026 – Mar 2026
Tech: React, Node.js, Express, MongoDB, Socket.io, FastAPI, scikit-learn, Groq Llama 3.3 70B
- Built a full-stack platform with swipe-based mutual matching, real-time Socket.io chat and JWT role-based access for founders, investors and admins (7 Mongoose models, 8 API route modules).
- Built a RAG pipeline (MongoDB retrieval + Groq Llama 3.3 70B) behind 5 GenAI tools (pitch analysis, investor memo, market analysis and more), grounding answers in live sector and competitor data.
- Designed a FastAPI microservice (TF-IDF, cosine similarity, Random Forest) for startup scoring and competitor detection: ~1.2 s per score, ~0.8 s per competitor search, mutual matches detected in under 500 ms.

Voice OS Lite – Offline AI Voice Assistant | Sep 2025 – Oct 2025
Tech: Python, Vosk, GPT4All (Llama 3.2 1B), Text-to-Speech
- Built a fully offline voice assistant (no internet or API keys) combining Vosk real-time speech-to-text with a local GPT4All LLM for question answering and spoken replies.
- Implemented voice-triggered actions (open Chrome, search files, read clipboard aloud, explain screenshots) in a modular design so new commands are easy to add.

Multilingual Medical Report Summarization System | Mar 2025 – Apr 2025
Tech: Python, Flask, spaCy, Tesseract OCR, MarianMT, DistilBART, ReportLab
- Built a 6-stage CPU-only NLP pipeline (OCR, language detection, translation, medical NER with negation handling, summarization, timeline) for PDF, image and text reports; cut processing from a 90 s cold start to 5–15 s using model warm-up and lazy loading.

EXPERIENCE

Machine Learning Intern | DHEE Centre of AI and Data Science | Jun 2024 – Jul 2024
- Built and trained a TensorFlow/Keras CNN (3 conv layers, 1.7M parameters) to classify 38 plant-disease classes across 14 crops on ~88K leaf images, reaching 97.3% validation accuracy (97.5% precision, 97.3% recall).
- Explored OpenCV preprocessing (histogram equalization, gamma correction) and built single-image disease prediction.

ACHIEVEMENTS AND CERTIFICATIONS

- HACKSTREET 3.0 Hackathon (Team Digital Dreamers): designed and built an education-accessibility prototype from concept to working demo within the hackathon timeframe.
- Participated in Dark-CTF 2026 (Crack-On Community), a cybersecurity capture-the-flag competition with 200+ teams.
- NPTEL: Affective Computing (Elite + Gold)
- NPTEL: Implementation of Human-Computer Interface (Elite)
- NPTEL: Distributed Systems (Elite + Silver)

LANGUAGES

English (Fluent) | Hindi (Native) | Japanese (Basic)